

Frequentist and Bayesian Cosmological Inference with the Pantheon+SH0ES Type Ia Supernova Sample

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Abstract

This report presents an end-to-end statistical analysis of the publicly released Pantheon+SH0ES Type Ia supernova compilation using the Hubble-flow subsample of $N = 1580$ supernovae selected by removing Cepheid-host calibrators and requiring $z_{\text{HD}} > 0.01$. The uploaded notebook performs all stages of the analysis: data loading, cosmological distance modelling, analytic marginalisation over the absolute-magnitude nuisance parameter M , frequentist profile-likelihood fitting, Bayesian Markov-chain Monte Carlo sampling, bootstrap and jackknife resampling, model comparison, and residual Gaussianity testing. For the flat Λ CDM model, the frequentist fit gives $\hat{\Omega}_m = 0.3491$ with 1σ Wilks interval $[0.3305, 0.3681]$ and $\chi^2_{\text{min}} = 694.15$. The Bayesian posterior gives $\Omega_m = 0.3490^{+0.0184}_{-0.0179}$, in close agreement with the frequentist profile likelihood. The two-parameter extensions provide only small improvements in χ^2 : $\Delta\chi^2 = 0.370$ for w CDM and $\Delta\chi^2 = 0.468$ for open Λ CDM relative to flat Λ CDM. Once parameter penalties are included, the information criteria and evidence calculations prefer flat Λ CDM: $\Delta\text{BIC} \simeq 6.99$ for w CDM and 6.90 for open Λ CDM, while the Savage-Dickey factors for the extended models relative to flat Λ CDM are 0.224 and 0.482, respectively. Residual diagnostics show that the normalized residuals have a nearly Gaussian shape when their empirical width is fitted, with KS $p = 0.0818$, but their standard deviation is only 0.6628, explaining the low reduced $\chi^2 \simeq 0.440$. The final conclusion from the notebook is that the supernova-only diagonal-error analysis supports flat Λ CDM over the tested one-parameter extensions, while also revealing that the diagonal error model is conservative.

1 Introduction

Type Ia supernovae remain a central observational probe of late-time cosmic expansion. The original supernova evidence for cosmic acceleration [1, 2] showed that the luminosity-distance-redshift relation cannot be explained by a simple decelerating matter-dominated universe. Modern compilations such as Pantheon+SH0ES [3, 4] provide large, well-calibrated samples that are suitable for comparing cosmological models and statistical inference frameworks.

The aim of the uploaded notebook, `Analysis.ipynb`, is not merely to fit a single cosmological parameter. It builds a complete analysis pipeline that compares frequentist and Bayesian methods on the same likelihood. The project studies three models:

- flat Λ CDM, with parameter Ω_m ;
- flat w CDM, with parameters (Ω_m, w) ;
- open Λ CDM, with parameters $(\Omega_m, \Omega_\Lambda)$ and $\Omega_k = 1 - \Omega_m - \Omega_\Lambda$.

The notebook then checks how stable the inferred parameters are under bootstrap resampling, how strongly the data prefer one model over another, and whether the Gaussian residual assumption is reasonable.

This revised report is written to match the actual notebook workflow and its numerical outputs. In particular, the original script-based description has been replaced with a notebook-based description, the author list has been corrected, the title is placed before the abstract, and figures are inserted close to the sections that discuss them.

2 Data and preprocessing

The notebook reads the official file `Pantheon+SH0ES.dat`. The full release contains 1701 light-curve entries. The analysis selects the Hubble-flow sample using two cuts:

1. remove Cepheid-host calibrators by requiring `IS_CALIBRATOR=0`;
2. require $z_{\text{HD}} > 0.01$ to reduce the effect of peculiar velocities at very low redshift.

After these cuts, the selected sample contains

$$N = 1580 \tag{1}$$

entries corresponding to 1466 unique supernova CIDs. The redshift range is

$$0.0102 \leq z_{\text{HD}} \leq 2.2614, \tag{2}$$

with median redshift 0.1852. The median diagonal magnitude uncertainty is 0.212 mag.

The relevant columns used by the notebook are z_{HD} , the corrected apparent magnitude $m_{B,\text{corr}}$, and the diagonal magnitude uncertainty $\sigma_{m_{B,\text{corr}}}^{\text{diag}}$. The analysis deliberately uses only the diagonal uncertainties and does not use the full covariance matrix. This choice is important because it later explains the low reduced χ^2 and the narrow normalized residual distribution.

3 Cosmological models and distance modulus

For a cosmological model with dimensionless expansion rate

$$E(z) = \frac{H(z)}{H_0}, \tag{3}$$

the line-of-sight comoving distance is

$$D_C(z) = \frac{c}{H_0} \int_0^z \frac{dz'}{E(z')}. \tag{4}$$

The luminosity distance is

$$d_L(z) = (1+z)D_M(z), \tag{5}$$

where D_M is the transverse comoving distance. For a flat universe $D_M = D_C$. For nonzero curvature, the notebook uses the standard sine or hyperbolic-sine curvature correction depending on the sign of Ω_k .

The theoretical distance modulus is

$$\mu(z) = 5 \log_{10} \left(\frac{d_L(z)}{\text{Mpc}} \right) + 25. \tag{6}$$

The notebook fixes a fiducial $h = 0.70$ inside the distance calculation. This does not remove the usual supernova degeneracy between H_0 and the absolute magnitude M because M is analytically profiled or marginalised.

The three expansion histories implemented in the code are

$$E_{\text{flat } \Lambda\text{CDM}}^2(z) = \Omega_m(1+z)^3 + (1 - \Omega_m), \quad (7)$$

$$E_{w\text{CDM}}^2(z) = \Omega_m(1+z)^3 + (1 - \Omega_m)(1+z)^{3(1+w)}, \quad (8)$$

$$E_{\text{open } \Lambda\text{CDM}}^2(z) = \Omega_m(1+z)^3 + \Omega_k(1+z)^2 + \Omega_\Lambda, \quad (9)$$

with

$$\Omega_k = 1 - \Omega_m - \Omega_\Lambda. \quad (10)$$

The luminosity-distance integral is evaluated numerically on an 8192-point grid and then interpolated.

4 Likelihood and analytic marginalisation over M

The observed effective distance modulus is modelled as

$$\mu_i^{\text{obs}} = m_{B,\text{corr}}^{(i)} - M. \quad (11)$$

Assuming independent Gaussian diagonal errors, the chi-squared function is

$$\chi^2(\theta, M) = \sum_{i=1}^N \frac{\left[m_{B,\text{corr}}^{(i)} - M - \mu(z_i; \theta) \right]^2}{\sigma_i^2}. \quad (12)$$

Here θ denotes the cosmological parameters and M is a nuisance parameter.

The notebook eliminates M analytically. Define

$$A = \sum_i \frac{\left[m_{B,\text{corr}}^{(i)} - \mu_i \right]^2}{\sigma_i^2}, \quad (13)$$

$$B = \sum_i \frac{m_{B,\text{corr}}^{(i)} - \mu_i}{\sigma_i^2}, \quad (14)$$

$$C = \sum_i \frac{1}{\sigma_i^2}. \quad (15)$$

The profiled chi-squared used in the code is

$$\chi_{\text{marg}}^2(\theta) = A - \frac{B^2}{C}. \quad (16)$$

The corresponding best-fit nuisance value is

$$\hat{M}(\theta) = \frac{B}{C}. \quad (17)$$

This is why the same numerical likelihood can be used for both the frequentist profile-likelihood analysis and the Bayesian posterior sampling with a flat prior on M .

5 Exploratory data analysis

The exploratory plot generated by the notebook is shown in Fig. 1. The top-left panel displays the Hubble diagram using the effective distance modulus $\mu_{\text{eff}} = m_{B,\text{corr}} - M$. The reference model in this figure uses $\Omega_m = 0.334$ and the fitted value $M = -19.3494$. The residuals relative to this reference have mean $+0.0073$ mag and standard deviation 0.1531 mag. The reduced chi-squared against this reference is 0.440 , already indicating that the diagonal error bars are conservative for this sample.

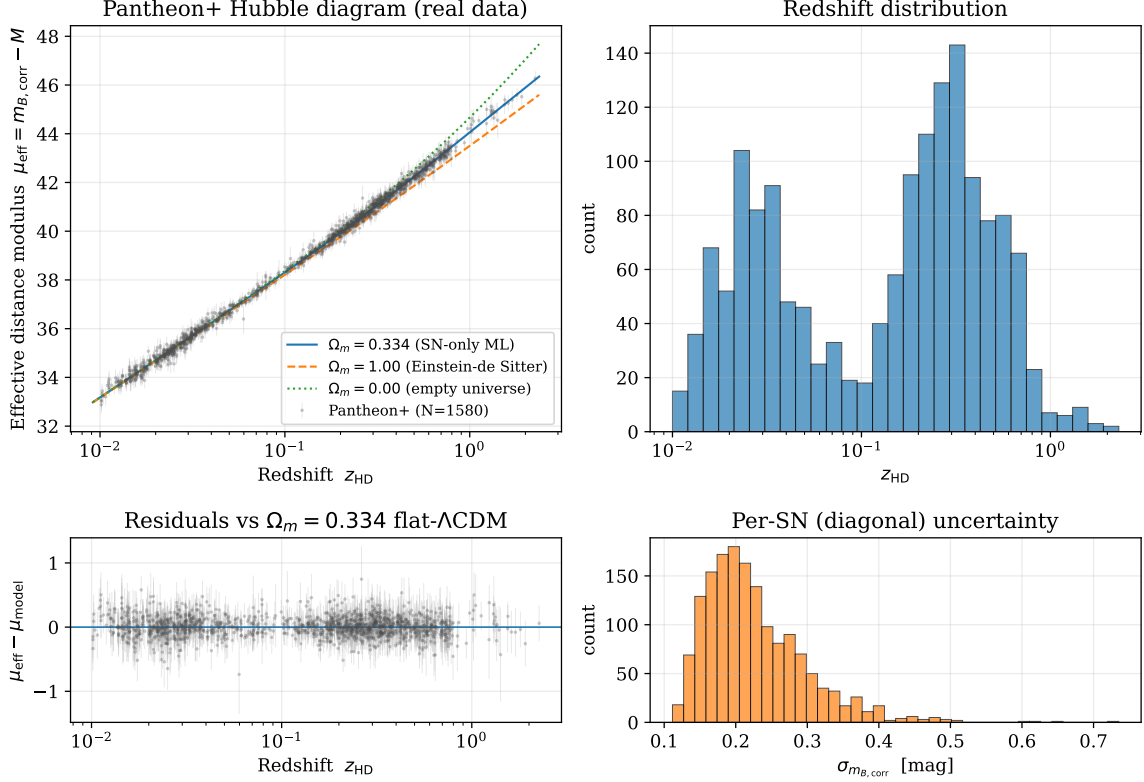


Figure 1: Exploratory Hubble-diagram diagnostics generated by the notebook. The panels show the effective distance modulus with reference flat- Λ CDM curves, the redshift distribution, residuals relative to the $\Omega_m = 0.334$ flat- Λ CDM reference, and the distribution of diagonal magnitude uncertainties.

6 Frequentist analysis

The notebook first fits the one-parameter flat Λ CDM model by minimizing Eq. (16) over Ω_m using a bounded scalar optimization. It obtains

$$\hat{\Omega}_m = 0.3491, \quad \chi_{\min}^2 = 694.15. \quad (18)$$

The profile-likelihood scan gives the following Wilks confidence intervals:

$$1\sigma : \Omega_m \in [0.3305, 0.3681], \quad (19)$$

$$2\sigma : \Omega_m \in [0.3123, 0.3876], \quad (20)$$

$$3\sigma : \Omega_m \in [0.2946, 0.4075]. \quad (21)$$

A quadratic fit to the profile near the minimum gives the Fisher estimate

$$\sigma_{\text{Fisher}} = 0.0188. \quad (22)$$

The notebook prints the frequentist reduced chi-squared using $N - 1$ in the profile-likelihood section, giving $694.15/1579 = 0.440$. In the residual diagnostics it uses $N - 2$ because both Ω_m and M have effectively been fitted, giving $694.15/1578 = 0.440$. The difference is numerically negligible, but the latter is the more appropriate goodness-of-fit degree-of-freedom count.

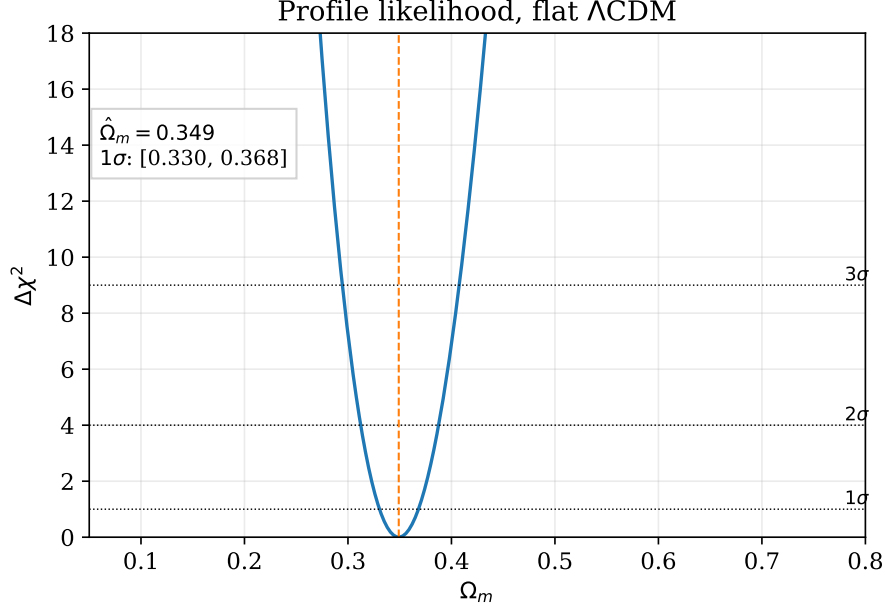


Figure 2: Flat- Λ CDM profile likelihood. The horizontal lines mark $\Delta\chi^2 = 1, 4, 9$, corresponding to nominal 1, 2, and 3σ confidence intervals for one fitted cosmological parameter.

The code then fits the two-parameter flat w CDM model using Nelder-Mead optimization. The maximum-likelihood point is

$$\hat{\Omega}_m = 0.2975, \quad \hat{w} = -0.8800, \quad \chi_{\min}^2 = 693.78. \quad (23)$$

Compared with flat Λ CDM, the improvement is only

$$\Delta\chi^2 = 694.146 - 693.776 = 0.370. \quad (24)$$

This is far too small to justify an extra parameter on its own. The contour plot in Fig. 3 also shows that $w = -1$ lies comfortably within the allowed region.

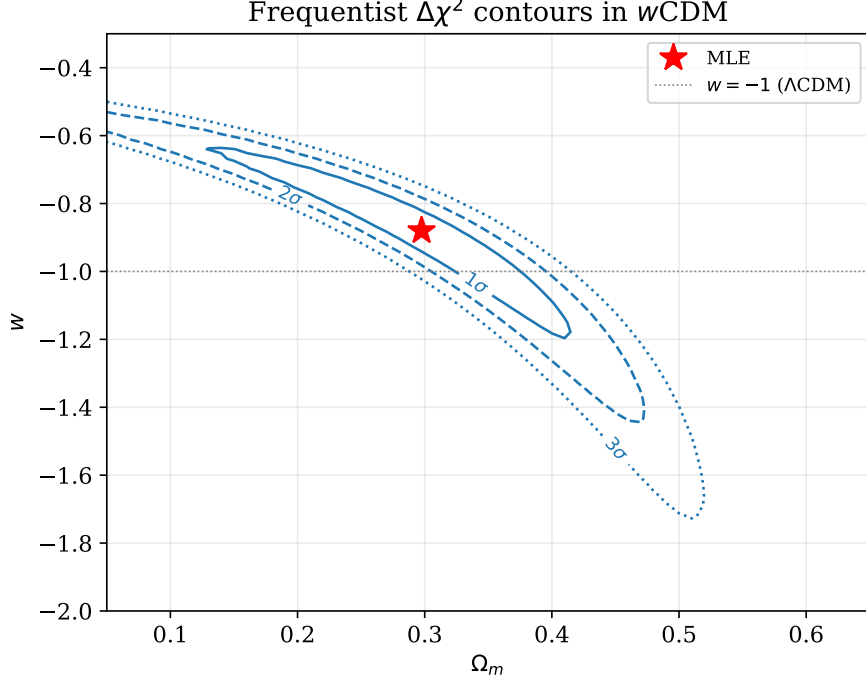


Figure 3: Frequentist $\Delta\chi^2$ contours in the flat w CDM model. The plotted contour levels are 2.30, 6.17, and 11.83, corresponding to nominal 1, 2, and 3σ regions for two parameters. The red star marks the maximum-likelihood estimate.

7 Bayesian MCMC analysis

The Bayesian part of the notebook uses the affine-invariant ensemble sampler `emcee` [5, 6]. The priors are uniform over the following ranges:

$$\Omega_m \sim \mathcal{U}(0, 1), \quad (25)$$

$$w \sim \mathcal{U}(-2.5, 0), \quad (26)$$

$$\Omega_\Lambda \sim \mathcal{U}(0, 1.5). \quad (27)$$

The flat Λ CDM chain uses 24 walkers, a 300-step initial burn-in, and 1500 production steps. After discarding 200 production steps and thinning by approximately half the measured autocorrelation time, the saved flat- Λ CDM chain has 2832 samples. The mean acceptance fraction is 0.814 and the reported autocorrelation time is $\tau = 22.84$.

For w CDM and open Λ CDM, the notebook uses 24 walkers, a 400-step initial burn-in, and 2500 production steps. The saved chains have 55200 samples each, with acceptance fractions 0.672 and 0.717, respectively.

For flat Λ CDM, the posterior result is

$$\Omega_m = 0.3490^{+0.0184}_{-0.0179}, \quad (28)$$

computed from the 15.865, 50, and 84.135 percentiles of the chain. Figure 4 compares the MCMC posterior with the normalized frequentist profile likelihood. Their agreement is expected because the likelihood is Gaussian-like and the prior is effectively flat over the posterior support.

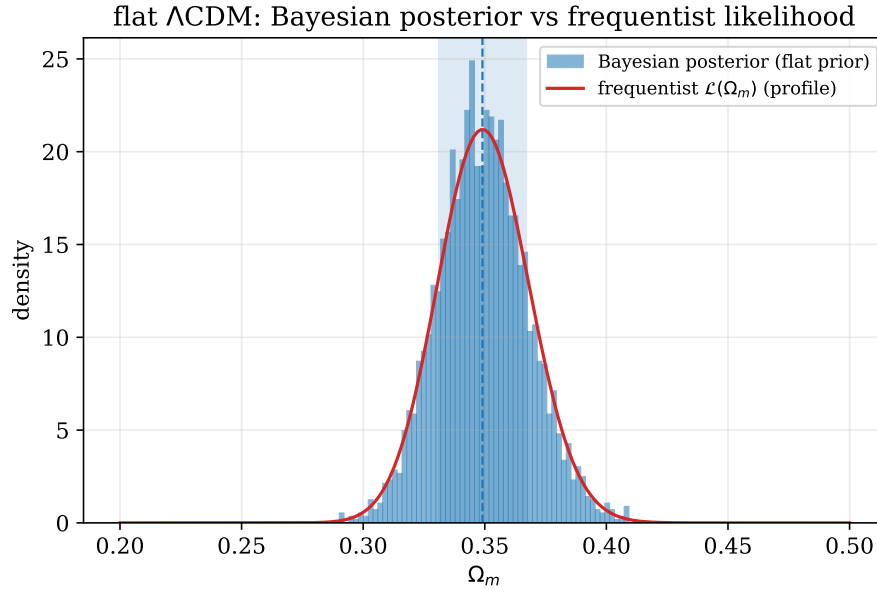


Figure 4: Flat- Λ CDM Bayesian posterior for Ω_m compared with the normalized frequentist profile likelihood. The two curves agree to within MCMC noise.

For the w CDM posterior, the notebook obtains

$$\Omega_m = 0.3127^{+0.0777}_{-0.1058}, \quad (29)$$

$$w = -0.9110^{+0.1830}_{-0.2014}. \quad (30)$$

This posterior is consistent with $w = -1$, so the data do not require a departure from a cosmological constant.

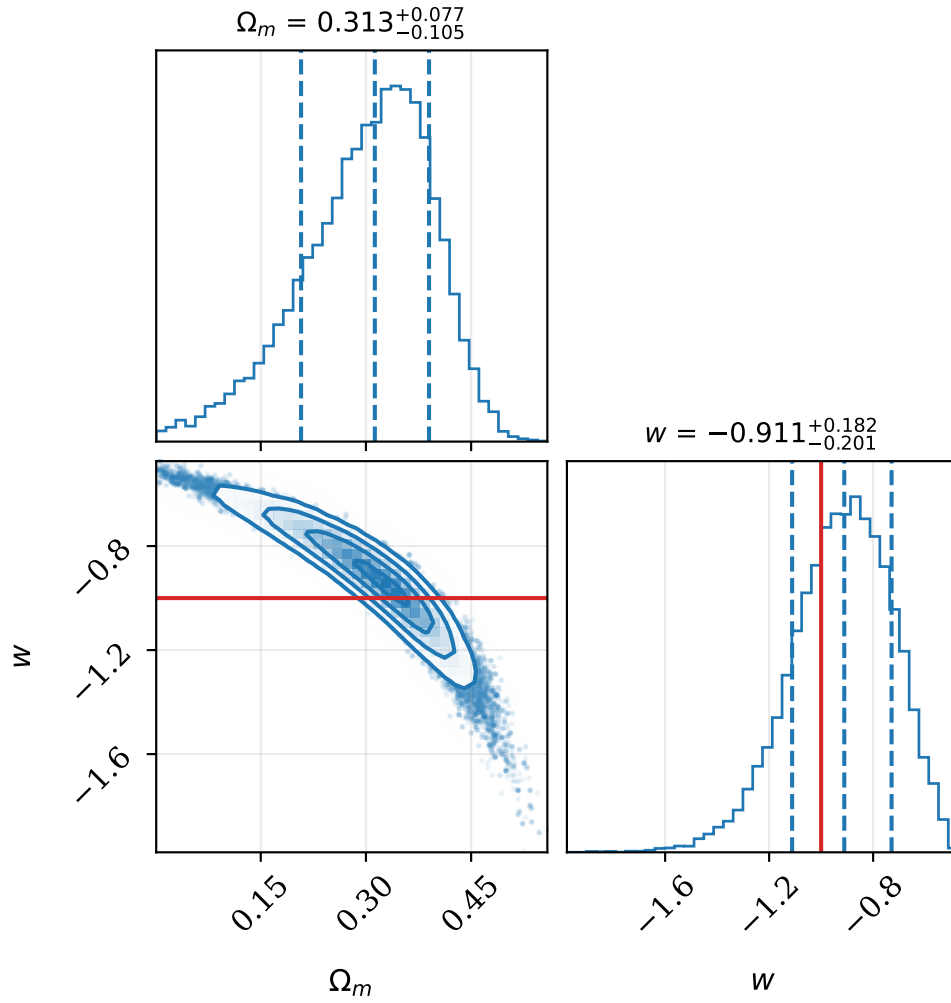


Figure 5: Bayesian corner plot for the flat w CDM model. The posterior includes $w = -1$ within the central credible region.

For open Λ CDM, the posterior summaries are

$$\Omega_m = 0.3016^{+0.0683}_{-0.0674}, \quad (31)$$

$$\Omega_\Lambda = 0.5855^{+0.0900}_{-0.0940}, \quad (32)$$

$$\Omega_k = 0.1132^{+0.1575}_{-0.1540}. \quad (33)$$

The curvature parameter is therefore consistent with $\Omega_k = 0$ at less than 1σ .

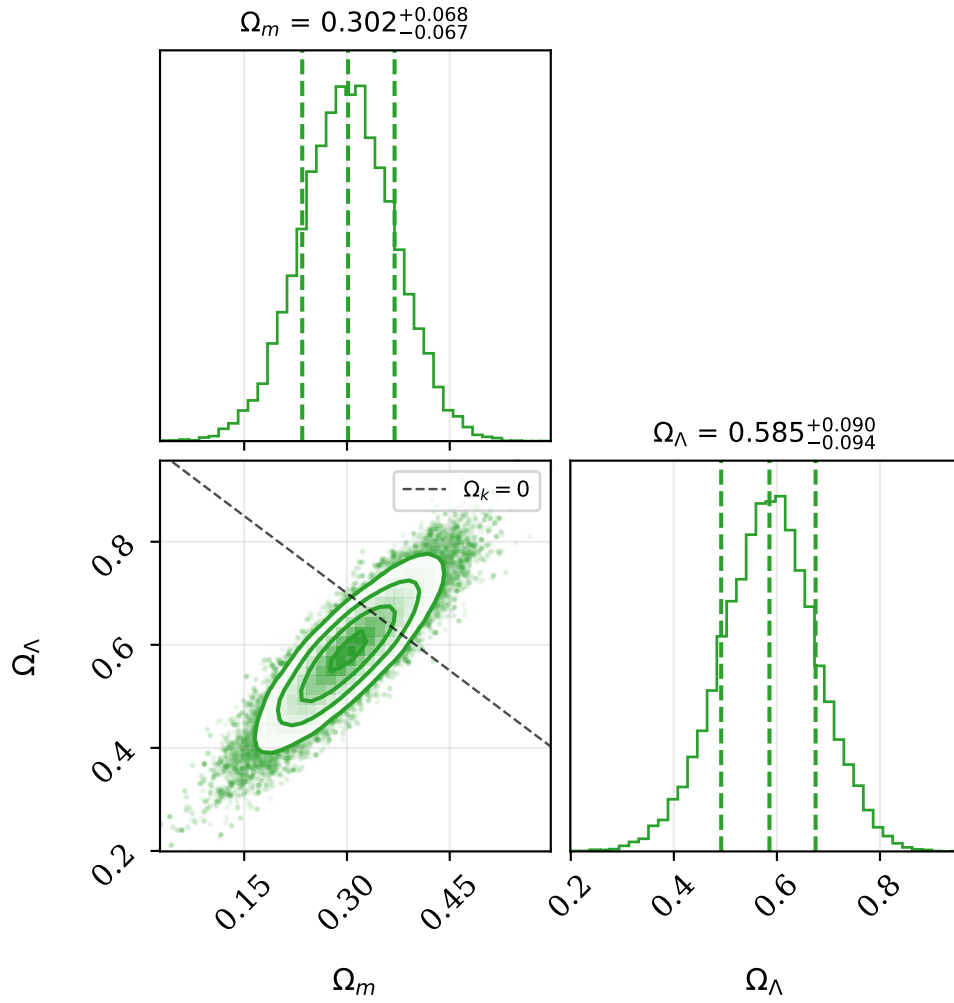


Figure 6: Bayesian corner plot for the open Λ CDM model. The dashed diagonal corresponds to $\Omega_m + \Omega_\Lambda = 1$, or equivalently $\Omega_k = 0$.

Finally, the notebook overlays Bayesian highest-posterior-density contours and frequentist $\Delta\chi^2$ contours for the w CDM model. As shown in Fig. 7, the two approaches give nearly the same central constraint region for this likelihood.

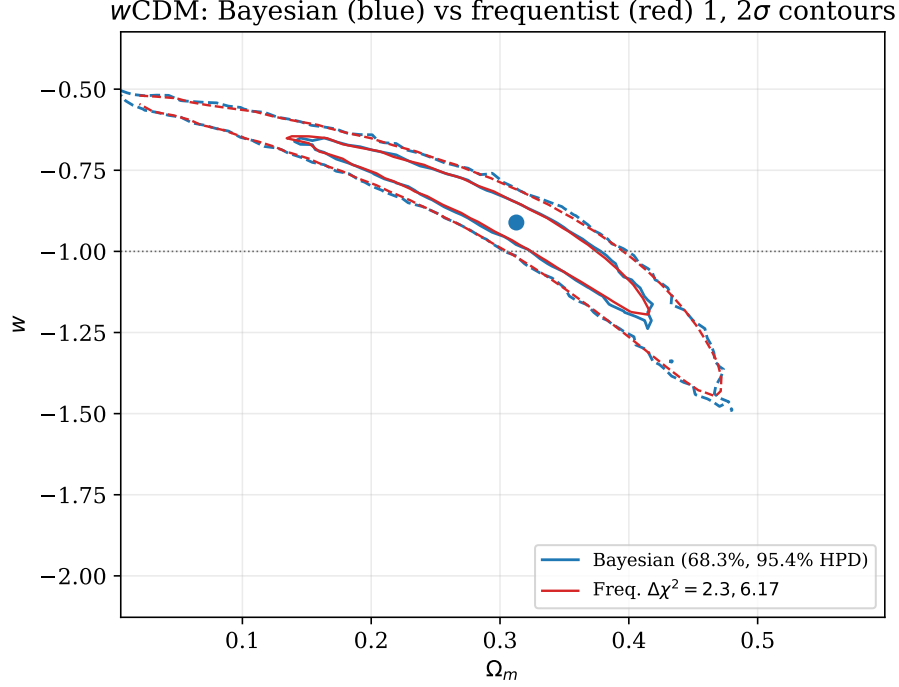


Figure 7: Direct comparison of Bayesian and frequentist constraints in the (Ω_m, w) plane. The central regions are nearly identical because the posterior is likelihood-dominated under the chosen flat priors.

8 Bootstrap and jackknife resampling

The notebook performs $B = 400$ non-parametric bootstrap resamples of the selected supernova triples $(z_i, m_{B,\text{corr}}^{(i)}, \sigma_i)$. For each bootstrap sample, it refits Ω_m under flat Λ CDM using the same marginalised chi-squared function.

The full-sample estimate is

$$\hat{\Omega}_{m\text{full}} = 0.34908. \quad (34)$$

The bootstrap distribution gives

$$\overline{\Omega}_{m\text{boot}} = 0.34998, \quad (35)$$

$$\sigma_{\text{boot}} = 0.01214, \quad (36)$$

$$\text{bias}_{\text{boot}} = +0.00091. \quad (37)$$

The percentile 68% interval is

$$[0.3392, 0.3613], \quad (38)$$

and the BCa 68% interval is

$$[0.3383, 0.3602]. \quad (39)$$

The BCa correction parameters printed by the notebook are

$$z_0 = -0.0313, \quad a = +0.0019. \quad (40)$$

The jackknife calculation in the notebook is explicitly thinned for speed. Since it does not use all N delete-one samples, its reported value $\sigma_{\text{jackknife}} \simeq 0.00551$ should be treated only as a diagnostic, not as a final uncertainty estimate.

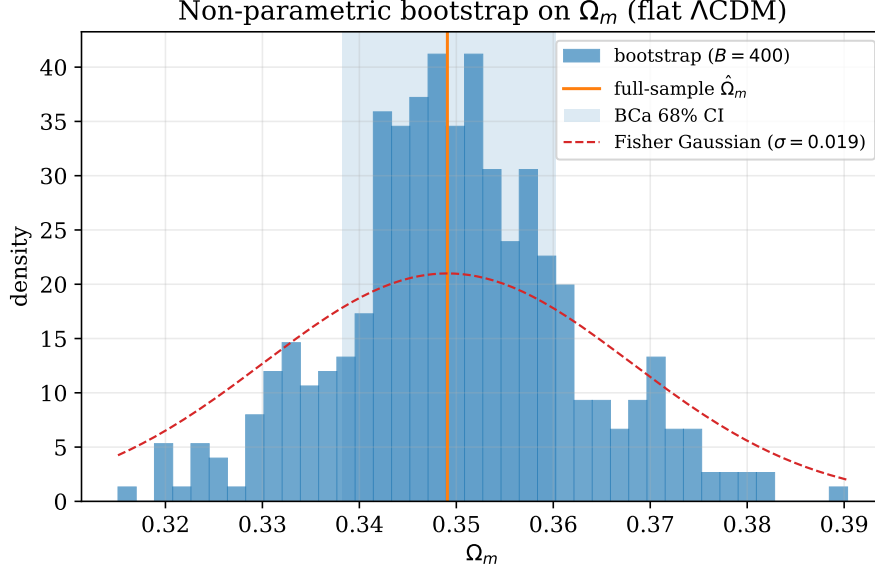


Figure 8: Bootstrap distribution of the flat- Λ CDM estimate of Ω_m . The bootstrap width is narrower than the Fisher width because the normalized residual scatter is below unity.

9 Model comparison

The notebook compares flat Λ CDM, flat w CDM, and open Λ CDM using AIC, BIC, Savage-Dickey density ratios, and a Laplace approximation to the Bayesian evidence. In the information criteria, the effective number of parameters includes the analytically eliminated nuisance parameter M . Thus

$$k_{\text{flat}} = 2, \quad k_{w\text{CDM}} = 3, \quad k_{\text{open}} = 3. \quad (41)$$

The information criteria are computed as

$$\text{AIC} = \chi_{\min}^2 + 2k, \quad (42)$$

$$\text{BIC} = \chi_{\min}^2 + k \ln N. \quad (43)$$

Table 1: Model-comparison results from the notebook. Differences are shown relative to flat Λ CDM, so positive ΔAIC and ΔBIC penalize the extended model relative to flat Λ CDM. The Savage-Dickey factor is reported as $B_{\text{extended,flat}}$, so values below 1 favor flat Λ CDM.

Model	k	$\chi_{\min}^2$	AIC	BIC	ΔAIC	ΔBIC
flat Λ CDM	2	694.146	698.15	708.88	0.00	0.00
w CDM	3	693.776	699.78	715.87	1.63	6.99
open Λ CDM	3	693.678	699.68	715.77	1.53	6.90

The extended models slightly lower χ^2 , but the improvement is too small to compensate for the extra parameter. The BIC penalty is especially strong because $\ln N \simeq 7.37$.

The Savage-Dickey calculations give

$$B_{w\text{CDM,flat}} = 0.2240, \quad (44)$$

$$B_{\text{open,flat}} = 0.4820. \quad (45)$$

Since both values are less than 1, they favor the nested flat Λ CDM model over the corresponding extension. Equivalently, the inverse factors are approximately 4.46 and 2.07 in favor of flat Λ CDM.

The Laplace log-evidences printed by the notebook are

$$\log Z_{\text{flat}} = -350.147, \quad (46)$$

$$\log Z_{w\text{CDM}} = -351.086, \quad (47)$$

$$\log Z_{\text{open}} = -351.247. \quad (48)$$

Therefore

$$\log B(\text{flat}/w\text{CDM}) = +0.939, \quad (49)$$

$$\log B(\text{flat}/\text{open}) = +1.100. \quad (50)$$

These values again prefer flat Λ CDM, but the evidence strength is modest rather than overwhelming.

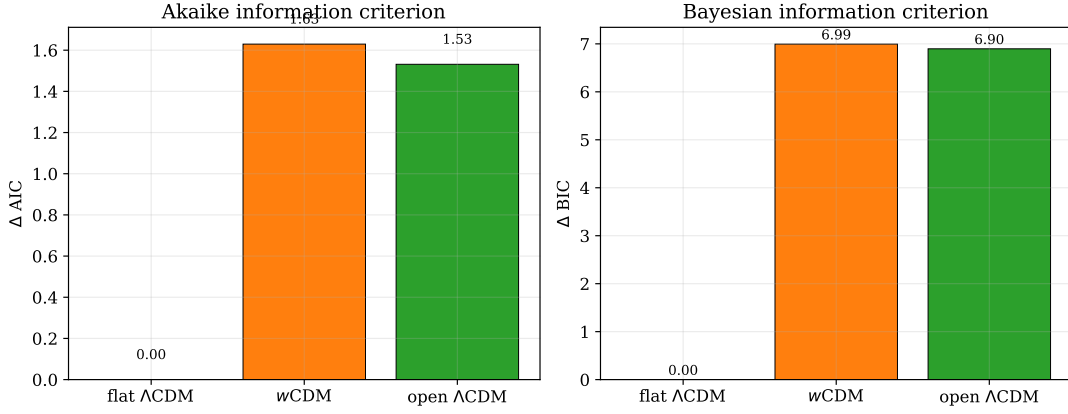


Figure 9: AIC and BIC differences generated by the notebook. Flat Λ CDM has the lowest AIC and BIC. BIC more strongly penalizes the extra parameter in w CDM and open Λ CDM.

10 Residual diagnostics and Gaussianity tests

The notebook computes normalized residuals using the flat- Λ CDM best fit:

$$r_i = \frac{\mu_i^{\text{obs}} - \mu_i^{\text{model}}}{\sigma_i}. \quad (51)$$

The residual summary is

$$\bar{r} = +0.0175, \quad (52)$$

$$s_r = 0.6628, \quad (53)$$

$$\text{skewness} = -0.0530, \quad (54)$$

$$\text{excess kurtosis} = +0.8857. \quad (55)$$

The chi-squared statistic is

$$\chi^2 = 694.15, \quad \nu = N - 2 = 1578, \quad \chi^2/\nu = 0.440. \quad (56)$$

The upper-tail chi-squared p -value is printed as approximately 1 because the observed χ^2 is much smaller than expected under correctly normalized independent errors. This should not

be read as a perfect fit. It is a warning that the diagonal uncertainties are too large for an independent-error interpretation.

Table 2: Residual Gaussianity tests from the notebook. The residuals are not consistent with a standard normal distribution, but they are close to Gaussian after fitting their mean and width.

Diagnostic	Notebook result
Shapiro-Wilk	$W = 0.9933$, $p = 1.372 \times 10^{-6}$
Anderson-Darling	$A^2 = 2.346$, rejected at all listed levels
KS vs $N(0, 1)$	$D = 0.1315$, $p = 2.833 \times 10^{-24}$
KS vs $N(\bar{r}, s_r)$	$D = 0.0317$, $p = 0.08184$
KL($p_{\text{emp}} N(0, 1)$)	0.1416 nats
Runs test	runs = 718, $E[R] = 790.1$, $z = -3.632$, $p = 2.81 \times 10^{-4}$

The key point is subtle but important. The residuals are not standard normal because their width is 0.6628, not 1. However, once the width is fitted, the KS test does not reject a Gaussian shape at standard significance levels. Thus the dominant issue is not a wildly non-Gaussian residual distribution. The dominant issue is that the diagonal errors are conservative when used without the full covariance structure.

Residual diagnostics, flat Λ CDM with $\hat{\Omega}_m = 0.349$

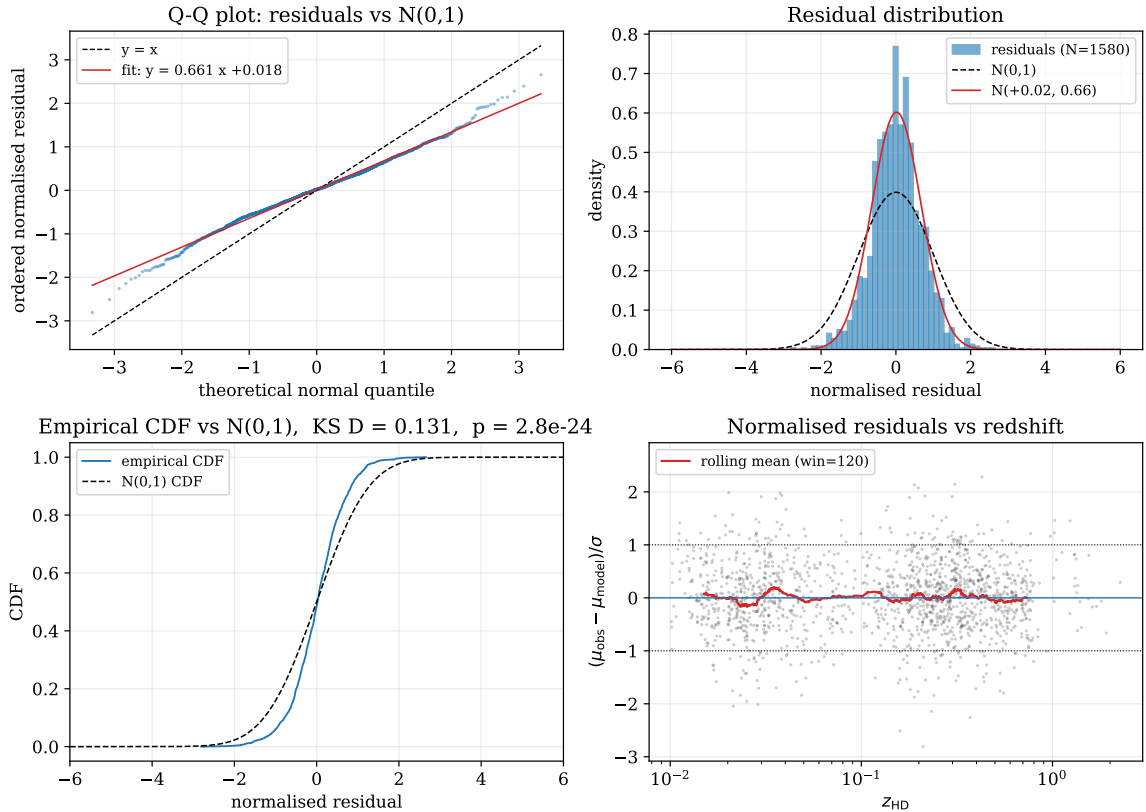


Figure 10: Residual diagnostics generated by the notebook. The Q-Q and histogram panels show a nearly Gaussian residual shape with sub-unit width. The empirical CDF strongly differs from $N(0, 1)$, while the residual-redshift panel shows mild redshift-dependent clustering of residual signs.

11 Discussion

The main statistical lesson from the notebook is that frequentist and Bayesian inference agree extremely well for the flat- Λ CDM one-parameter problem. This is not magic. The likelihood is smooth and close to Gaussian near its maximum, the prior is flat across the posterior region, and the nuisance parameter M is linear enough to be eliminated analytically. Under these conditions, the Bayesian posterior is essentially the normalized frequentist likelihood.

The same practical agreement persists in the central region of the w CDM two-parameter constraint. The Bayesian and frequentist contours differ only in the tails, where prior volume, finite chain sampling, and non-Gaussian boundary effects become more relevant.

Model comparison tells a slightly different story from pure best-fit optimization. Both w CDM and open Λ CDM reduce χ^2 by less than 0.5, but each adds one cosmological parameter. AIC is mildly in favor of flat Λ CDM, BIC is more clearly in favor of flat Λ CDM, and both Savage-Dickey and Laplace evidence calculations also favor the simpler model. The clean conclusion is that the tested extensions are not warranted by this diagonal-error supernova-only analysis.

The residual diagnostics are also essential. A reduced χ^2 of 0.440 is not a medal. It means the normalized residuals are too narrow. The notebook shows this directly: the residual standard deviation is 0.6628. This is consistent with using only diagonal statistical uncertainties from a dataset whose full treatment requires correlated covariance terms.

12 Conclusions

The uploaded notebook supports the following final conclusions:

- The selected Hubble-flow sample contains $N = 1580$ supernova entries after removing calibrators and applying $z_{\text{HD}} > 0.01$.
- The flat- Λ CDM frequentist estimate is $\hat{\Omega}_m = 0.3491$ with Wilks 1σ interval $[0.3305, 0.3681]$.
- The flat- Λ CDM Bayesian result is $\Omega_m = 0.3490^{+0.0184}_{-0.0179}$, essentially identical to the frequentist result.
- The w CDM maximum-likelihood point is $(\Omega_m, w) = (0.2975, -0.8800)$, but the improvement over flat Λ CDM is only $\Delta\chi^2 = 0.370$.
- The open- Λ CDM maximum-likelihood point is $(\Omega_m, \Omega_\Lambda) = (0.3036, 0.5877)$, with $\Delta\chi^2 = 0.468$ relative to flat Λ CDM.
- AIC, BIC, Savage-Dickey ratios, and Laplace evidence all prefer flat Λ CDM over the two tested one-parameter extensions.
- Bootstrap resampling gives $\sigma_{\text{boot}} = 0.01214$ and BCa 68% interval $[0.3383, 0.3602]$ for Ω_m .
- The normalized residuals are close to Gaussian in shape after fitting their empirical width, but their standard deviation is 0.6628, so the diagonal errors are conservative under an independent-error interpretation.

Overall, the notebook implements a coherent frequentist and Bayesian cosmological inference pipeline. The report-worthy result is not simply that $\Omega_m \simeq 0.35$. The sharper conclusion is that, for this exact code and diagonal-error treatment, flat Λ CDM is the statistically preferred model, while the residual diagnostics clearly warn that the uncertainty model is conservative and incomplete without the full covariance matrix.

13 Code and Data Availability

The complete analysis code used in this project is available at:

https://github.com/AS-28980/Data_Science_Project

The Pantheon+SH0ES dataset used for the supernova and Cepheid-based cosmological analysis is publicly available through the official data release repository:

<https://github.com/PantheonPlusSH0ES/DataRelease>

All results, plots, and numerical estimates reported in this work were generated using the analysis notebook and the publicly released Pantheon+SH0ES data products.

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